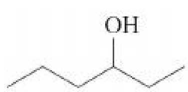
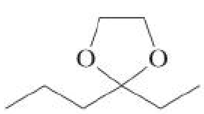
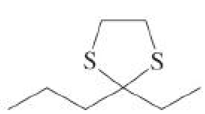
the spectra in terms of the structures.

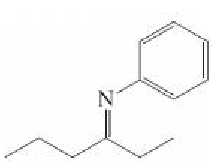
31. Following are spectroscopic and analytical characteristics for an unknown substance. Propose a structure. Empirical formula: C8H16O. $C_8H_{16}O$. 1H NMR: δ=0.90 (t, 3 H), ^1H NMR : $\delta = 0.90$ (t, 3 H), 1.0–1.6 (m, 8 H), 1.0–1.6 (m, 8 H), 2.05 (s, 3 H), 2.05 (s, 3 H), 2.25 (t, 2 H) ppm; 2.25 (t, 2 H) ppm; 13C NMR ^{13}C NMR seven signals between δ=14.0 $\delta = 14.0$ and and 43.8 ppm, 43.8 ppm, and one at 208.9 ppm; 208.9 ppm; IR: 1715 cm−1. 1715 cm^{-1}. UV: λmax(ε)=280(15) nm; UV : $\lambda_{max}(\acute{E}) = 280(15)$ nm; MS: m/z=128 (M+·); MS : $m/z = 128\ (M^{+\cdot})$; intensity of (M+1)+ $(M + 1)^+$ peak is 9% 9% of M+· $M^{+\cdot}$ peak; important fragments are at m/z=113 (M-15)+, $m/z = 113\ (M - 15)^+$, m/z=85 (M-43)+, $m/z = 85\ (M - 43)^+$, m/z=71 (M-57)+, $m/z = 71\ (M - 57)^+$, m/z=58 (M-70)+ $m/z = 58\ (M - 70)^+$ (the second largest peak), and m/z=43 (M-85)+ $m/z = 43\ (M - 85)^+$ (the base peak).

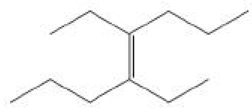
32. Reaction review. Without consulting the Reaction Road Map on pp. 850–851, suggest reagents to convert each of the starting materials below into 3-hexanone.

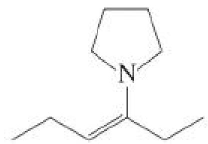
a. 

b. 

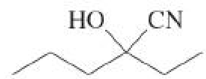
c. 

d. 

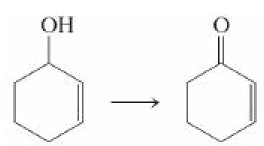
e. 

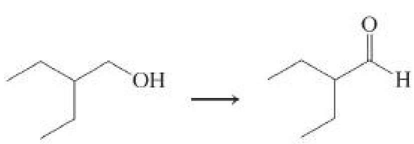
f. 

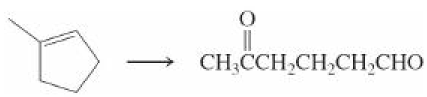
g. 

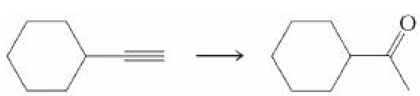
h. 

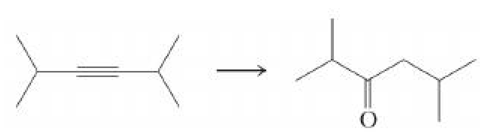
33. Indicate which reagent or combination of reagents is best suited for each of the following reactions.

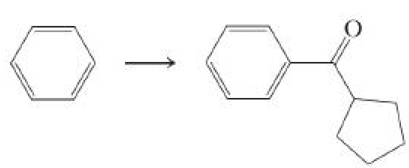
a. 

b. 

c. 

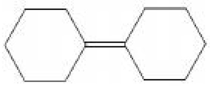
d. 

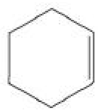
e. 

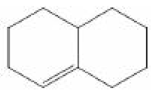
f. 

34. Write the expected products of ozonolysis (Section 12-12) of each of the following molecules.

a. CH3CH2CH2CH=CH2 $CH_3CH_2CH_2CH$â CH_2

b. 

c. 

d. 

35. For each of the following groups, rank the molecules in decreasing order of reactivity toward addition of a nucleophile to the most electrophilic sp2 sp^2-hybridized carbon.

a. $(CH_3)_2C{=}O$, $(CH_3)_2C{=}NH$, $(CH_3)_2C{=}\overset{+}{O}H$

b. $CH_3\overset{O}{\overset{\|}{C}}CH_3$, $CH_3\overset{O}{\overset{\|}{C}}\overset{O}{\overset{\|}{C}}CH_3$, $CH_3\overset{O}{\overset{\|}{C}}\overset{O}{\overset{\|}{C}}\overset{O}{\overset{\|}{C}}CH_3$

c. BrCH2COCH3, CH3COCH3, CH3CHO, BrCH2CHO
$BrCH_2COCH_3$, CH_3COCH_3, CH_3CHO, $BrCH_2CHO$

36. Give the expected products of reaction of butanal with each of the following reagents.

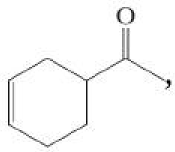
a. LiAlH4, (CH3CH2)2O, $LiAlH_4$, $(CH_3CH_2)_2O$, then H+, H2O H^+, H_2O

b. CH3CH2MgBr, (CH3CH2)2O, CH_3CH_2MgBr, $(CH_3CH_2)_2O$, then H+, H2O H^+, H_2O

c. HOCH2CH2OH, H+ $HOCH_2CH_2OH$, H^+

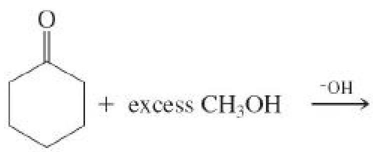
37. Give the expected products of reaction of 2-pentanone with each of the reagents in Problem 36.

38. Give the expected products of reaction of 1-(3-cyclohexenyl)-1-

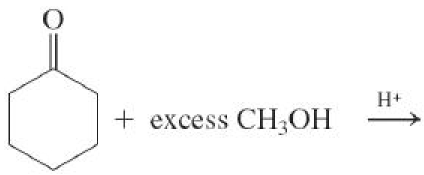
ethanone, , with each of the reagents in Problem 36.

39. Give the expected product(s) of each of the following reactions.

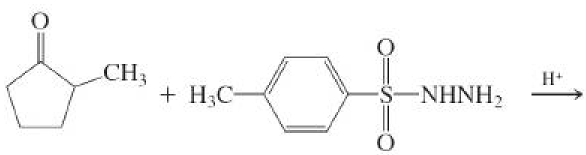
a.



b.

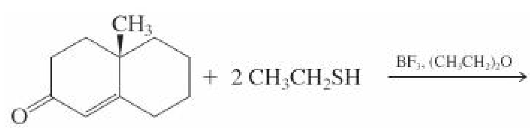


c.

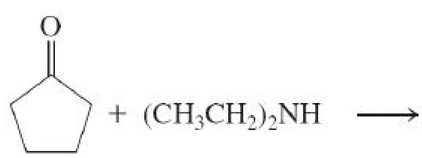


d. $CH_3\overset{\overset{O}{\|}}{C}CH_3 + HOCH_2\overset{\overset{OH}{|}}{C}HCH_2CH_2CH_3 \xrightarrow{H^+}$

e.



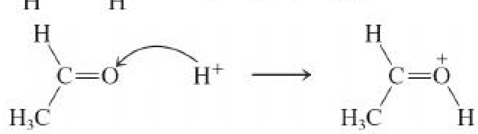
f.

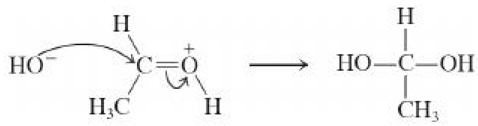


40. The equations for the mechanism of hydration of acetaldehyde under acidic aqueous conditions below are flawed. **(a)** For each step, add the missing electron pairs and identify the problem(s). **(b)** Write a correct step-by-step mechanistic description of the process.

1. $H{-}O{-}H \longrightarrow H^+ + {}^-OH$

2.



3. 

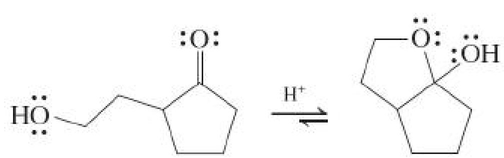
41. Formulate detailed mechanisms for **(a)** the formation of the hemiacetal of acetaldehyde and methanol under both acid- and base-catalyzed conditions; **(b)** the formation of the intramolecular hemiacetal of 5-hydroxypentanal ([Section 17-7](#)), again under both acid- and base-catalyzed conditions, and **(c)** the BF3BF_3-catalyzed reaction of CH3SH CH_3SH with butanal to give a thioacetal ([Section 17-8](#)).

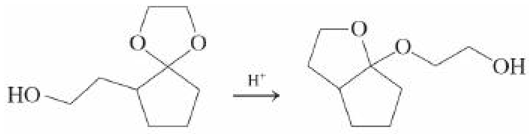
42. **CHALLENGE** Overoxidation of primary alcohols to carboxylic acids is caused by the water present in the usual aqueous acidic Cr(VI) reagents. The water adds to the initial aldehyde product to form a hydrate, which is further oxidized ([Sections 17-4](#) and [17-6](#)). In view of these facts, explain the following two observations. **(a)** Water adds to ketones to form hydrates, but no overoxidation follows the conversion of a secondary alcohol into a ketone. **(b)** Successful oxidation of primary alcohols to aldehydes by the waterfree PCC reagent requires that the alcohol be added slowly to the Cr(VI)$Cr(VI)$ reagent. If, instead, the PCC is added to the alcohol, (a so-called "inverse addition" procedure), a new side reaction forms an ester. This result is illustrated for 1-butanol.

$$CH_3CH_2CH_2CH_2OH \xrightarrow[\text{Inverse addition}]{\text{PPC, } CH_2Cl_2} CH_3CH_2CH_2\overset{\overset{\displaystyle O}{\|}}{C}OCH_2CH_2CH_2CH_3$$

(c) Give the products expected from reaction of 3-phenyl-1-propanol and water-free CrO3CrO_3 (1) when the alcohol is added to the oxidizing agent and (2) when the oxidant is added to the alcohol (inverse addition).

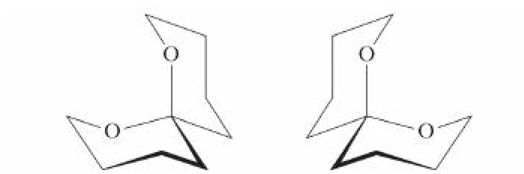
43. Explain the results of the following reactions by means of mechanisms.

a. 

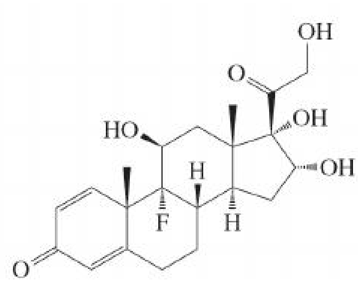
b. 

c. Explain why hemiacetal formation may be catalyzed by either acid or base, but acetal formation is catalyzed only by acid, not by base.

44. The two isomeric structures below are naturally occurring insect pheromones. The isomer on the left attracts the male olive fruit fly; the one on the right, the female. **(a)** What kind of isomeric relationship exists between these two structures? **(b)** What functional group is contained in these molecules? **(c)** Both of these compounds are hydrolyzed under aqueous acidic conditions. Draw the products. Are the product molecules from the two starting isomers the same or different?

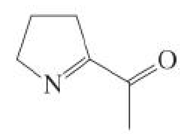


45. The biological activity of fluorinated organics has been widely exploited in the development of pharmaceutical agents (Real Life 6-1). The compound below is a precursor to the synthesis of a very promising anti-inflammatory drug that, as of late 2016, was in final testing for FDA approval as a time-release treatment for osteoarthritis. This affliction causes severe joint pain that can be difficult to manage effectively.



The actual drug candidate is prepared by reaction of this molecule with acetone in the presence of catalytic acid. Given the functional groups in this molecule and the transformations of ketones introduced in this chapter, speculate on the possible product(s) of this process and write mechanisms for their formation.

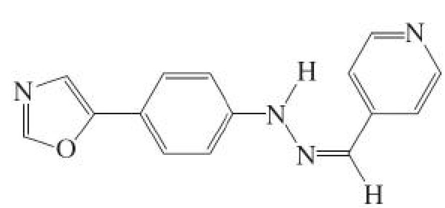
46. The molecule below, 2-acetyl-1-pyrroline (2-AP), is associated with the aroma and flavor of white bread and basmati rice. In higher concentrations, it gives cooked popcorn its characteristic buttery aroma and flavor. Its odor is very intense: We can detect 2-AP in water at concentrations below one nanogram/liter. The buttery aspect derives from a hydrolysis reaction. Identify the functional group sensitive to hydrolysis, write out a hydrolysis mechanism under acidic conditions, and give the structure of the product.



2-Acetyl-1-pyrroline

The urine of bearcats contains 2-AP. It is thought that they use the molecule as a sex attractant (a pheromone, Real Life 2-2 and Section 12-17). So if anyone asks you why bearcats smell like popcorn. . .

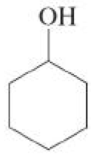
47. Neurodegenerative diseases, such as Alzheimer's and Parkinson's, are of particular concern due to their devastating effects on the quality of human life. Extensive efforts have been underway for decades to find working treatments. In recent years several small molecules have shown promising results in animal cells, although as yet unsolved technical problems prevent their testing in humans. One of these molecules is the hydrazine shown here, code named "compound B."

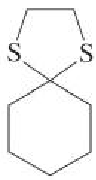


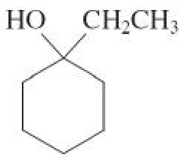
"Compound B"

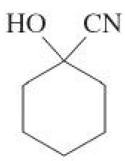
From what two compounds may compound B be prepared? Show the mechanism of that reaction.

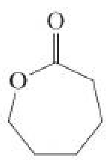
48. Reaction review II. Without consulting the Reaction Road Map on p. 852, suggest reagents to convert cyclohexanone into each of the compounds below.

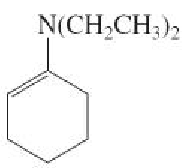
a. 

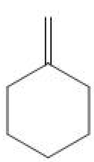
b. 

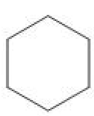
c. 

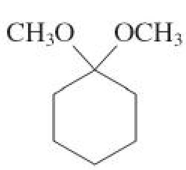
d. 

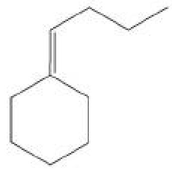
e. 

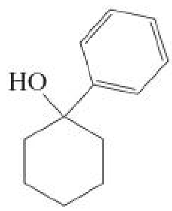
f. 

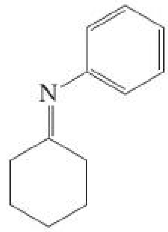
g. 

h. 

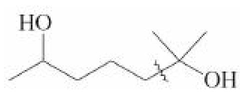
i. 

j. 

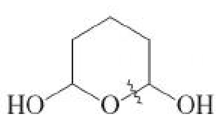
k. 

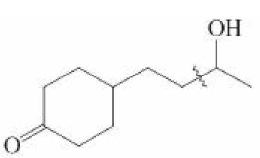
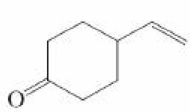
l. 

49. Propose reasonable syntheses of each of the following molecules, beginning with the indicated starting material. [**Hint:** Work backward by considering the strategic retrosynthetic C–CC– C disconnections indicated (〰), and then analyze for the potential need of protecting groups.]

a.  from 

b. $C_6H_5N{=}C(CH_2CH_3)_2$ from 3-pentanol

c.  from 1,5-pentanediol

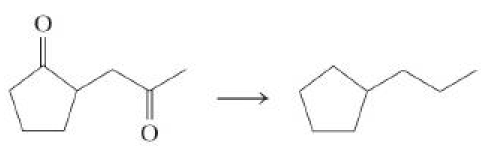
d.  from 

50. The UV absorptions and colors of 2,4-dinitrophenylhydrazone derivatives of aldehydes and ketones depend sensitively on the structure of the carbonyl compound. Suppose that you are asked to identify the contents of three bottles whose labels have fallen off. The labels indicate that one bottle contained butanal, one contained *trans*-2-butenal, and one contained *trans*-3-phenyl-2-propenal. The 2,4-dinitrophenylhydrazones prepared from the contents of the bottles have the following characteristics.

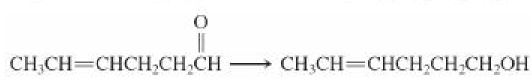
Bottle 1: m.p. 187-188°C; λmax=377 nm; $\text{m.p. } 187-188^{\circ}\text{C}; \lambda_{\max} = 377 \text{ nm};$ orange color
Bottle 2: m.p. 121-122°C; λmax=358 nm; $\text{m.p. } 121-122^{\circ}\text{C}; \lambda_{\max} = 358 \text{ nm};$ yellow color
Bottle 3: m.p. 252-253°C; λmax=394 nm; $\text{m.p. } 252-253^{\circ}\text{C}; \lambda_{\max} = 394 \text{ nm};$ red color

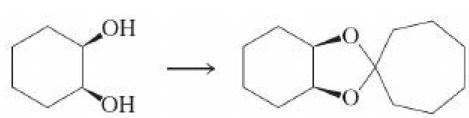
Match up the hydrazones with the aldehydes (*without* first looking up the melting points of these derivatives), and explain your choices. (**Hint:** See Section 14-11.)

51. Indicate the reagent(s) best suited to effect these transformations.

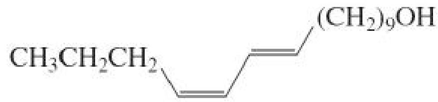
a. 

b. 

c. 

d. 

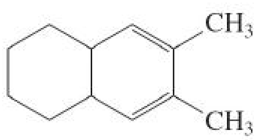
52. The molecule bombykol, whose structure is shown on top of the right column, is a powerful insect pheromone, the sex attractant of the female silk moth (see Section 12-17). It was initially isolated in the amount of 12 mg from extraction of 2 tons of moth pupae. Propose a synthesis from BrCH2(CH2)9OH$BrCH_2(CH_2)_9OH$ and CH3CH2CH2C≡CCHO, $CH_3CH_2CH_2C\equiv CCHO$, using a Wittig reaction (which in this system is known to form a trans double bond) as one of the steps of your sequence.



Bombykol

53. For each of the following molecules, propose *two* methods of synthesis from the different precursor molecules indicated.

a. CH3CH═CHCH2CH(CH3)2CH_3CHâ€$CHCH_2CH(CH_3)_2$ from (1) an aldehyde and (2) a different aldehyde

b.  from (1) a dialdehyde and (2) a diketone

54. Three isomeric ketones with the molecular formula C7H14O$C_7H_{14}O$ are converted into heptane by Clemmensen reduction. Compound A gives a single product upon Baeyer-Villiger oxidation; compound B gives two different products in very different yields; compound C gives two different products in virtually a 1 : 1 ratio. Identify A, B, and C.

55. Give the product(s) of reaction of hexanal with each of the following reagents.

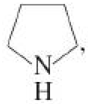
a. HOCH2CH2OH,×H+$HOCH_2CH_2OH,\ H^+$

b. LiAlH4,$LiAlH_4$, then H+,×H2O$H^+,\ H_2O$

c. NH2OH,×H+$NH_2OH,\ H^+$

d. NH2NH2,×KOH,$NH_2NH_2,\ KOH,$ heat

e. (CH3)2CHCH2CH≡P(C6H5)3$(CH_3)_2CHCH_2CH{\equiv}P(C_6H_5)_3$

f. , H^+

g. Ag+,×NH3,×H2O$Ag^+,\ NH_3,\ H_2O$

h. CrO3,×H2SO4,×H2O$CrO_3,\ H_2SO_4,\ H_2O$

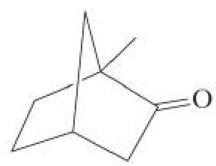
i. HCNHCN

56. Give the product(s) of reaction of cycloheptanone with each of the reagents in Problem 55.

57. Formulate in full detail the mechanism for the Wolff-Kishner reduction of 1-phenylethanone (acetophenone) to ethylbenzene (see p. 835).

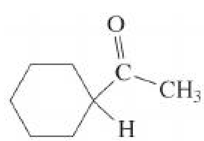
58. The general equation for the Baeyer-Villiger oxidation (see p. 840) begins with a reaction between a ketone and a peroxycarboxylic acid to form a peroxy analog of a hemiacetal. Formulate a detailed mechanism for this process.

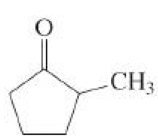
59. **(a)** Formulate a detailed mechanism for the Baeyer-Villiger oxidation of the ketone shown below (refer to Exercise 17-23).

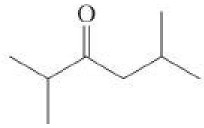


(b) Under Baeyer-Villiger conditions, aldehydes convert to carboxylic acids—for example, benzaldehyde goes to benzoic acid. Explain.

60. Give the two theoretically possible Baeyer-Villiger products from each of the following compounds. Indicate which one is formed preferentially.

a. 

b. 

c. 

d.